

The Dirichlet Pólya Problem for Spherical Shells in General Dimension: Exact Dimension Lifting and the Shifted-Tail Bottleneck

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Abstract

For a spherical shell in dimension $d \geq 3$, the separated radial problem has Bessel orders $\ell + (d-2)/2$ and spherical-harmonic multiplicities $\kappa_{d,\ell}$. We give an exact branching decomposition of the strict shell-action proxy into positive integer- and half-integer shifted planar tails. We also prove a signed primitive comparison and retain its full positive remainder. This yields the exact identity

$$W_d - P_d^< = \mathcal{B}_d(A) + \sum_{m \geq 0} \binom{m+d-3}{d-3} D_A \left(m + \frac{d-2}{2} \right).$$

Consequently, one dimension-free positive shifted-tail inequality implies the sharp Dirichlet Pólya inequality for every spherical shell and every $d \geq 3$. The dimension lift is proved here; the inner shifted-tail inequality remains the central open lemma. We prove the outer and terminal tail regions and reduce the first interface-cell milestone to a finite list of one-variable ball-action walls. No assertion in this manuscript treats the remaining shifted-tail lemma as proved.

1 Statement and status

For $0 < \rho < 1$ let

$$A_{\rho,1}^{(d)} = \{x \in \mathbb{R}^d : \rho < |x| < 1\}, \quad d \geq 3,$$

and use the strict counting convention

$$N_D(A_{\rho,1}^{(d)}, K^2) := \#\{j : \lambda_j^D(A_{\rho,1}^{(d)}) < K^2\}.$$

Write

$$\omega_d = \frac{\pi^{d/2}}{\Gamma(d/2 + 1)}, \quad L_d = \frac{\omega_d}{(2\pi)^d}.$$

The desired estimate is

$$N_D(A_{\rho,1}^{(d)}, K^2) \leq L_d |A_{\rho,1}^{(d)}| K^d = \frac{1 - \rho^d}{2^d \Gamma(d/2 + 1)^2} K^d. \quad (1)$$

For $a > 0$, define the zero-extended ball action

$$G_a(z) = \begin{cases} \frac{1}{\pi} \left(\sqrt{a^2 - z^2} - z \arccos \frac{z}{a} \right), & 0 \leq z \leq a, \\ 0, & z > a, \end{cases} \quad (2)$$

put $\mu = \rho K$, and set

$$A(z) = A_{\rho,K}(z) := G_K(z) - G_\mu(z). \quad (3)$$

For $x > 0$, let

$$[x]_< := \#\{n \in \mathbb{N} : n < x\} = \max\{0, [x] - 1\}.$$

In particular, $[n]_< = n - 1$ at a positive integral wall.

The only unproved statement on the primary path is the following.

Conjecture 1.1 (Positive shifted shell tails). *For $0 < \rho < 1$, $K > 0$, and*

$$r \in \tfrac{1}{2}\mathbb{N},$$

one has

$$T_A(r) := [A(r) + \tfrac{1}{4}]_< + 2 \sum_{j \geq 1} [A(r+j) + \tfrac{1}{4}]_< \leq 2 \int_r^K A(z) \, dz. \quad (4)$$

The zero extension makes the sum finite and makes the assertion trivial when $r \geq K$.

Sections 2–5 prove that Conjecture 1.1 implies (1) in every dimension $d \geq 3$. Section 6 records the proved tail regions and a sharper reduction of the first interface-cell problem.

2 General-dimensional spectral proxy

Put

$$a_d = \frac{d-2}{2}, \quad \nu_{\ell,d} = \ell + a_d.$$

The degree- ℓ spherical harmonics on $\mathbb{S}^{d-1}$ have multiplicity

$$\kappa_{d,\ell} = \binom{\ell+d-1}{d-1} - \binom{\ell+d-3}{d-1} = \frac{(2\ell+d-2)(\ell+d-3)!}{\ell!(d-2)!}. \quad (5)$$

Here and below a binomial coefficient with lower index larger than its nonnegative upper index is zero.

Under the unitary radial substitution

$$u(r) = r^{(d-1)/2} R(r),$$

the degree- ℓ block becomes

$$L_{\ell,d} = -\frac{d^2}{dr^2} + \frac{\nu_{\ell,d}^2 - \frac{1}{4}}{r^2} \quad \text{on } (\rho, 1), \quad u(\rho) = u(1) = 0. \quad (6)$$

Its positive eigenfrequencies are the zeros of

$$D_{\nu,\rho}(k) = J_\nu(\rho k) Y_\nu(k) - Y_\nu(\rho k) J_\nu(k), \quad \nu = \nu_{\ell,d}.$$

With the standard continuous Bessel phase, let

$$\gamma_{\rho,K}(\nu) = \frac{\theta_\nu(K) - \theta_\nu(\rho K)}{\pi}.$$

The determinant factorization and phase monotonicity give the exact strict count

$$N_D(A_{\rho,1}^{(d)}, K^2) = \sum_{\ell \geq 0} \kappa_{d,\ell} [\gamma_{\rho,K}(\nu_{\ell,d})]_{<.} \quad (7)$$

The real-order phase-difference enclosure of [2, 3] gives, for $0 \leq \nu \leq K$,

$$\gamma_{\rho,K}(\nu) < A_{\rho,K}(\nu) + \frac{1}{4}. \quad (8)$$

For $\nu \geq K$, both the exact count and the action proxy vanish. Since $x < y$ implies $[x]_{<} \leq [y]_{<}$, including when y is integral, (7)–(8) imply the wall-safe proxy bound

$$N_D(A_{\rho,1}^{(d)}, K^2) \leq P_d^<(\rho, K), \quad P_d^< := \sum_{\ell \geq 0} \kappa_{d,\ell} q_\ell, \quad (9)$$

where

$$q_\ell = [A(\ell + a_d) + \frac{1}{4}]_{<}. \quad (10)$$

The argument above is dimension-independent after the substitutions $\nu = \ell + a_d$ and (5). In particular, the published phase enclosure is already a real-order result: no interpolation is introduced when d changes parity.

3 Exact harmonic branching

Let $p = d - 2$, so $a_d = p/2$, and define

$$r_m = m + \frac{p}{2}, \quad c_{d,m} = \binom{m + p - 1}{p - 1} = \binom{m + d - 3}{d - 3}. \quad (11)$$

Proposition 3.1 (Exact shifted-tail decomposition). *For every finitely supported sequence $(q_\ell)_{\ell \geq 0}$,*

$$\sum_{\ell \geq 0} \kappa_{d,\ell} q_\ell = \sum_{m \geq 0} c_{d,m} \left(q_m + 2 \sum_{j \geq 1} q_{m+j} \right). \quad (12)$$

Consequently, for (10),

$$\boxed{P_d^< = \sum_{m \geq 0} c_{d,m} T_A(r_m)}. \quad (13)$$

This identity is exact at every strict action wall.

Proof. The generating functions are

$$\sum_{\ell \geq 0} \kappa_{d,\ell} s^\ell = \frac{1 + s}{(1 - s)^{d-1}}, \quad \sum_{m \geq 0} c_{d,m} s^m = \frac{1}{(1 - s)^{d-2}},$$

and

$$1 + 2 \sum_{j \geq 1} s^j = \frac{1 + s}{1 - s}.$$

Coefficient comparison gives

$$\kappa_{d,\ell} = c_{d,\ell} + 2 \sum_{m=0}^{\ell-1} c_{d,m}.$$

Substitution and finite reindexing prove (12). For (10), one has $q_{m+j} = [A(r_m + j) + 1/4]_{<}$, which proves (13). No floor has been approximated or moved. $\square$

The shifts are

$$r_m \in \begin{cases} \mathbb{N}_0 + \frac{1}{2}, & d \text{ odd}, \\ \mathbb{N}, & d \text{ even}. \end{cases}$$

Thus Conjecture 1.1 is asked only on the two grids that actually occur.

4 The signed Weyl primitive

Define the right-continuous branching staircase

$$S_d(z) = \sum_{\substack{m \geq 0 \\ r_m \leq z}} c_{d,m} \quad (14)$$

and its discrepancy primitive

$$\Delta_d(z) = \int_0^z \left(S_d(u) - \frac{u^p}{p!} \right) du. \quad (15)$$

Proposition 4.1 (Signed branching primitive). *For every $d \geq 3$ and $z \geq 0$,*

$$\boxed{\Delta_d(z) \leq 0.} \quad (16)$$

For $d = 3$, equality at positive z occurs precisely at the positive integers. For $d \geq 4$, the inequality is strict for $z > 0$.

Proof. For $0 \leq z < r_0 = p/2$, $S_d(z) = 0$, so

$$\Delta_d(z) = -\frac{z^{p+1}}{(p+1)!} \leq 0.$$

Let $M \geq 0$ and $r_M = M + p/2$. The hockey-stick identity gives

$$\int_0^{r_M} S_d(u) du = \sum_{m=0}^{M-1} (r_M - r_m) c_{d,m} = \binom{M+p}{p+1}.$$

Therefore

$$(p+1)! \Delta_d(r_M) = \prod_{j=0}^p (M+j) - \left(M + \frac{p}{2} \right)^{p+1} \leq 0, \quad (17)$$

because the arithmetic mean of $M, M+1, \dots, M+p$ is $M + p/2$.

On the half-open cell $[r_M, r_{M+1})$,

$$S_d(z) = \binom{M+p}{p}, \quad \Delta'_d(z) = \binom{M+p}{p} - \frac{z^p}{p!}.$$

If the critical point lies in this cell, it is

$$z_M^* = \left(\prod_{j=1}^p (M+j) \right)^{1/p}.$$

Direct substitution into the cell formula gives

$$(p+1)! \Delta_d(z_M^*) = p(z_M^*)^p \left(z_M^* - M - \frac{p+1}{2} \right) \leq 0, \quad (18)$$

since z_M^* is the geometric mean of $M+1, \dots, M+p$ and their arithmetic mean is $M+(p+1)/2$. If the critical point is outside the cell, the cell maximum is at an endpoint, already covered by (17). This proves (16).

For $p=1$, the cell maximum is $z=M+1$ and has value zero. For $p \geq 2$, the consecutive factors in both AM–GM applications are unequal, which gives the stated strictness. $\square$

5 Exact defect identity and conditional closure

Define

$$D_A(r) = 2 \int_r^K A(z) dz - T_A(r) \quad (19)$$

and the continuous payment

$$W_d = \frac{2}{p!} \int_0^K z^p A(z) dz. \quad (20)$$

Lemma 5.1 (Exact Weyl normalization). *For $p = d-2$,*

$$\int_0^a z^p G_a(z) dz = \frac{\Gamma((p+1)/2)}{4\sqrt{\pi}(p+2)\Gamma((p+4)/2)} a^{p+2}. \quad (21)$$

Consequently,

$$W_d = \frac{1 - \rho^d}{2^d \Gamma(d/2 + 1)^2} K^d = L_d |A_{\rho,1}^{(d)}| K^d. \quad (22)$$

Proof. Scale $z = ax$ in (2), integrate the term containing $\arccos x$ by parts, and evaluate the resulting beta integrals. This gives (21). Multiplying the difference of (21) at $a = K$ and $a = \rho K$ by $2/p!$, the duplication formula for Γ reduces the coefficient to

$$\frac{1}{2^{p+2} \Gamma((p+4)/2)^2} = \frac{1}{2^d \Gamma(d/2 + 1)^2}.$$

The last equality in (22) follows from $|A_{\rho,1}^{(d)}| = \omega_d(1 - \rho^d)$. $\square$

Proposition 5.2 (Branching-defect identity). *Let*

$$\mathcal{B}_d(A) = 2 \int_0^K (-A'(z))(-\Delta_d(z)) dz. \quad (23)$$

Then $\mathcal{B}_d(A) \geq 0$ and

$$W_d - P_d^< = \mathcal{B}_d(A) + \sum_{m \geq 0} c_{d,m} D_A(r_m). \quad (24)$$

Proof. The shell action is absolutely continuous, decreasing, and vanishes at K . Fubini gives

$$\sum_{m \geq 0} c_{d,m} 2 \int_{r_m}^K A(z) dz = 2 \int_0^K S_d(z) A(z) dz.$$

Since $\Delta'_d = S_d - z^p/p!$ almost everywhere and both endpoint terms vanish, integration by parts gives

$$2 \int_0^K S_d A = W_d + 2 \int_0^K A \Delta'_d = W_d - 2 \int_0^K A' \Delta_d = W_d - \mathcal{B}_d(A).$$

Both A' and Δ_d are nonpositive, so $\mathcal{B}_d(A) \geq 0$. Subtract (13) and use (19). $\square$

Theorem 5.3 (Conditional all-dimensional shell theorem). *If Conjecture 1.1 holds, then (1) holds for every $d \geq 3$, $0 < \rho < 1$, and $K \geq 0$. It then also holds for every shell $A_{r,R}^{(d)} = \{x \in \mathbb{R}^d : r < |x| < R\}$ and every spectral parameter.*

Proof. The case $K = 0$ is trivial, so assume $K > 0$. Every r_m in (11) lies in $\frac{1}{2}\mathbb{N}$, so the conjecture gives $D_A(r_m) \geq 0$. Equations (9), (24), and (22) yield

$$N_D(A_{\rho,1}^{(d)}, K^2) \leq P_d^< \leq W_d = L_d |A_{\rho,1}^{(d)}| K^d.$$

For a shell with outer radius R , take $\rho = r/R$ and $K = R\sqrt{\Lambda}$ and use Dirichlet dilation. The non-strict counting form follows by applying the strict estimate at $\Lambda + \varepsilon$ and letting $\varepsilon \downarrow 0$. $\square$

6 Proved tail regions and the first interface cell

The action derivative is

$$-A'(z) = \frac{1}{\pi} \begin{cases} \arccos \frac{z}{K} - \arccos \frac{z}{\mu}, & 0 < z < \mu, \\ \arccos \frac{z}{K}, & \mu \leq z < K. \end{cases} \quad (25)$$

Thus A is decreasing and concave on $(0, \mu)$, decreasing and convex on (μ, K) , and A, A' match at μ .

Proposition 6.1 (Outer and terminal tails). *Conjecture 1.1 holds in either of the following cases:*

1. $r \geq \mu$;
2. $G_K(r) \leq 3/4$.

Proof. If $r \geq \mu$, then $A = G_K$ on $[r, K]$; the claim is trivial when $r \geq K$. Otherwise set $B = \lceil K - r \rceil$ and zero-extend

$$g(s) = G_K(r + s) \quad (0 \leq s \leq K - r)$$

to the integer interval $[0, B]$. Since $G_K(K) = G'_K(K) = 0$, this extension is nonnegative, decreasing, convex, $1/2$ -Lipschitz, and vanishes at B . The convex shifted-floor theorem of [1] therefore bounds its ordinary-floor tail by $2 \int_0^B g$. At every sample,

$$[g(j) + 1/4]_< \leq \lfloor g(j) + 1/4 \rfloor.$$

The strict tail is consequently bounded by $2 \int_0^B g = 2 \int_r^K G_K$.

If $G_K(r) \leq 3/4$, monotonicity and $A \leq G_K$ imply $A(r + j) + 1/4 \leq 1$ for every $j \geq 0$. Every strict bracket is therefore zero. Notice that equality at 1 is harmless precisely because the bracket is strict. $\square$

We next sharpen the prescribed first milestone. For the ball action set

$$T_K(r) = [G_K(r) + \tfrac{1}{4}]_< + 2 \sum_{j \geq 1} [G_K(r + j) + \tfrac{1}{4}]_<, \quad D_K(r) = 2 \int_r^K G_K(z) dz - T_K(r). \quad (26)$$

The wall-safe translation in the preceding proof gives $D_K(r) \geq 0$.

Lemma 6.2 (A low-action $1/3$ -Lipschitz reserve). *Let g be strictly decreasing and convex on $[0, b]$, with $g(b) = 0$ and $\text{Lip}(g) \leq 1/3$. If*

$$\frac{3}{4} < g(0) < 1,$$

then

$$2 \int_0^b g(z) \, dz - \left([g(0) + \tfrac{1}{4}]_{<} + 2 \sum_{j \geq 1} [g(j) + \tfrac{1}{4}]_{<} \right) > \frac{11}{16}. \quad (27)$$

The function is understood to be zero-extended after b .

Proof. Let $h = g(0)$ and let $x \in (0, b)$ be defined by $g(x) = 3/4$. Since $g(j) < 1$, a sample contributes exactly one if and only if $j < x$. Therefore the bracketed count is

$$2[x] - 1.$$

The Lipschitz bound and $g(b) = 0$ give

$$g(z) \geq \max\{h - z/3, 0\}, \quad 2 \int_0^b g(z) \, dz \geq 3h^2 > \frac{27}{16}.$$

If $x \leq 1$, the count is one and (27) follows.

Suppose $x > 1$ and put $a = (h - 3/4)/x$. Convexity implies that, for $z \geq x$,

$$g(z) \geq \max\left\{\frac{3}{4} - a(z - x), 0\right\}.$$

Hence

$$\int_0^b g(z) \, dz \geq \frac{3}{4}x + \frac{(3/4)^2}{2a} = x \left(\frac{3}{4} + \frac{9}{32(h - 3/4)} \right) > \frac{15}{8}x.$$

Since $2[x] - 1 < 2x + 1$, the defect is larger than

$$\frac{15}{4}x - (2x + 1) = \frac{7}{4}x - 1 > \frac{3}{4} > \frac{11}{16}.$$

□

Proposition 6.3 (Exact one-interface reduction). *Assume*

$$0 < \mu - r < 1, \quad r \in \tfrac{1}{2}\mathbb{N}.$$

Define

$$n_0 = [G_K(r) + \tfrac{1}{4}]_{<} - [G_K(r) - G_\mu(r) + \tfrac{1}{4}]_{<} \in \mathbb{N}_0 \quad (28)$$

and

$$\delta_\mu(r) = \int_r^\mu G_\mu(z) \, dz.$$

Then

$$\boxed{D_A(r) = D_K(r) + n_0 - 2\delta_\mu(r)}. \quad (29)$$

Moreover,

$$2\delta_\mu(r) < \frac{4(\mu - r)^{5/2}}{15\sqrt{\mu}} < \frac{4\sqrt{2}}{15} < 1. \quad (30)$$

Consequently, the shifted-tail inequality holds if any of the following is true:

1. $n_0 \geq 1$;
2. $G_K(\max\{r, K/2\}) \geq 1$;
3. $r \geq K/2$.

The only remaining one-interface configurations satisfy

$$\boxed{r < \frac{K}{2}, \quad K < \frac{1}{\omega_0}, \quad n_0 = 0, \quad \omega_0 := G_1(1/2) = \frac{\sqrt{3}}{2\pi} - \frac{1}{6}.} \quad (31)$$

In particular,

$$r \in \left\{ \frac{1}{2}, 1, \frac{3}{2}, 2, \frac{5}{2}, 3, \frac{7}{2}, 4, \frac{9}{2} \right\}.$$

Proof. Since $r + j > \mu$ for every $j \geq 1$, all samples after the first agree with the ball samples. Thus $T_A(r) = T_K(r) - n_0$. Subtracting $2 \int_r^\mu G_\mu$ from the ball integral proves (29).

The turning estimate

$$G_\mu(z) < \frac{(\mu - z)^{3/2}}{3\sqrt{\mu}} \quad (0 < z < \mu)$$

integrates to (30), using $\mu > r \geq 1/2$ and $\mu - r < 1$. If $n_0 \geq 1$, then $D_K(r) \geq 0$ and (29) is positive.

For $r < K$, use the same integer endpoint $B = \lceil K - r \rceil$ and zero-extended function $g(s) = G_K(r + s)$. Put

$$s_* = \max\{0, K/2 - r\}.$$

Then g is $1/3$ -Lipschitz on $[s_*, B]$ and $g(s_*) = G_K(\max\{r, K/2\})$. The refined ordinary-floor theorem of [3, Theorem 3.4], followed by $\lceil \cdot \rceil < \lfloor \cdot \rfloor$, gives

$$D_K(r) \geq \frac{1}{2} \lfloor G_K(\max\{r, K/2\}) \rfloor. \quad (32)$$

For $r \geq K$ the same inequality is trivial. Hence the second condition and (30) prove the claim.

If $r \geq K/2$ and $G_K(r) < 1$, the translated ball action is globally $1/3$ -Lipschitz. Its tail is either zero or Lemma 6.2 applies. In the latter case $D_K(r) > 11/16 > 4\sqrt{2}/15$, so (29) is again positive. This proves the third condition.

If none of the three conditions holds, then $r < K/2$ and $G_K(K/2) = \omega_0 K < 1$, giving (31). The elementary rational bounds

$$\frac{265}{153} < \sqrt{3} < \frac{26}{15}, \quad \frac{333}{106} < \pi < \frac{355}{113}$$

(the square-root bounds follow by squaring) give $1/10 < \omega_0 < 1/9$. Consequently

$$\frac{9}{2} < \frac{1}{2\omega_0} < 5.$$

Since $r < K/2 < 1/(2\omega_0)$ and $r \in \frac{1}{2}\mathbb{N}$, only the nine displayed shifts remain. $\square$

Remark 6.4 (What remains in the first milestone). Proposition 6.3 does not finish the one-interface theorem. It replaces a two-parameter fractional-interface problem by a bounded nine-shift residual family: $2r < K < 1/\omega_0$, and $r < \mu < \min\{r + 1, K\}$, with the explicit no-loss condition $n_0 = 0$. The required inequality there is exactly

$$D_K(r) \geq 2\delta_\mu(r).$$

This is the next scalar reserve lemma to prove.

Proposition 6.5 (Finite ball-wall reduction). *In the residual family of Proposition 6.3, put*

$$B = G_K(r) + \frac{1}{4}, \quad q = [B]_<, \quad e = B - q \in (0, 1],$$

and set

$$E_r(K) = D_K(r) - e.$$

Then

$$2\delta_\mu(r) < e. \tag{33}$$

Consequently, the residual one-interface inequality follows from $E_r(K) \geq 0$.

On every open K -interval containing no ball-sample wall, E_r is strictly increasing. At a wall of the first sample $G_K(r) + 1/4$, the two unit drops in $D_K(r)$ and e cancel, so E_r is continuous. At a wall of a later sample $G_K(r+j) + 1/4$, $j \geq 1$, E_r drops by two for each crossed level. It is therefore enough to check the right-hand limits at the finitely many later-sample walls, together with the first positive-count wall and any lower endpoint $K = 2r$ at which the count is already positive.

More explicitly, all walls that can occur have the form

$$G_K(r+j) = n - \frac{1}{4}, \quad j \geq 0, \quad n \in \{1, 2, 3\}, \quad r+j < 10. \tag{34}$$

Proof. Let $h = G_\mu(r)$. The strict remainder convention gives the exact equivalence

$$n_0 = 0 \iff [B-h]_< = [B]_< \iff h < e.$$

In particular, equality $h = e$ causes a strict-count drop and is not part of the no-loss case. The function G_μ is strictly convex on $[r, \mu]$, with endpoint values h and 0. It lies strictly below its endpoint chord, and therefore

$$2\delta_\mu(r) < h(\mu - r) < e,$$

which proves (33). If $T_K(r) = 0$, then $T_A(r) = 0$ and the desired inequality is immediate. Otherwise it is enough, by (29), to prove $D_K(r) \geq e$.

Away from sample walls the integer-valued tail and q are constant. Writing $x = r/K$, differentiation of the ball action and its tail integral gives

$$\pi E'_r(K) = K \left(\arccos x - x\sqrt{1-x^2} \right) - \sqrt{1-x^2}, \tag{35}$$

and

$$\pi E''_r(K) = \arccos x + x\sqrt{1-x^2} - \frac{x^2}{K\sqrt{1-x^2}}. \tag{36}$$

Here $x < 1/2$ and $K > 1$. Hence

$$\pi E''_r(K) > \frac{\pi}{3} - \frac{1}{2\sqrt{3}} > 0.$$

For $r \geq 1$, evaluation at $K = 2r$ gives

$$\pi E'_r(2r) = \frac{2\pi r}{3} - \frac{\sqrt{3}}{2}(r+1) > 0.$$

For $r = 1/2$, positivity of the tail count implies $G_K(r) > 3/4$. Since G_K is $1/2$ -Lipschitz and vanishes at K , this forces $K > 2$. Formula (35) is already positive at $(r, K) = (1/2, 2)$, since

$$\pi E'_{1/2}(2) = 2 \arccos \frac{1}{4} - \frac{3\sqrt{15}}{8} > \frac{2\pi}{3} - \frac{3}{2} > 0.$$

Equation (36) then keeps it positive. Thus E_r is strictly increasing between walls.

The stated jump rules follow directly from the weights $1, 2, 2, \dots$ in $T_K(r)$ and from the reset of e from 1 to 0 at a first-sample wall. Finally, $K < 1/\omega_0 < 10$ and $G_K(z) < K/\pi < 10/3$. Thus only $n = 1, 2, 3$ and samples $r + j < 10$ can occur, proving the finite reduction. $\square$

7 Cell recurrence and the corrected global target

For $x_j = r + j$, set

$$Q_j = [A(x_j) + \frac{1}{4}]_<, \quad D_j = 2 \int_{x_j}^K A(z) dz - \left(Q_j + 2 \sum_{k>j} Q_k \right).$$

Then

$$D_j - D_{j+1} = 2 \int_{x_j}^{x_{j+1}} A(z) dz - Q_j - Q_{j+1}. \quad (37)$$

If

$$e_j = A(x_j) + \frac{1}{4} - Q_j \in (0, 1]$$

and

$$C_j = 2 \int_{x_j}^{x_{j+1}} A(z) dz - A(x_j) - A(x_{j+1}),$$

then

$$D_j - D_{j+1} = C_j - \frac{1}{2} + e_j + e_{j+1}. \quad (38)$$

On a fully inner cell,

$$C_j = - \int_0^1 u(1-u) A''(x_j + u) du \geq 0.$$

Individual increments in (38) can nevertheless be negative, and a negative increment need not be paid by the next floor-drop cell. Thus the viable block formulation must retain the terminal convex reserve. If J is the first index with $x_J \geq \mu$, the exact target is

$$\boxed{D_0 = \sum_{j=0}^{J-1} (D_j - D_{j+1}) + D_J \geq 0,} \quad (39)$$

where $D_J \geq 0$ is the translated convex ball-tail defect. A proof of (39) needs a quantitative, not merely nonnegative, lower bound for D_J or an equivalent matched shelf reserve.

8 A dimension-dependent no-mode region

The radial form (6), the one-dimensional Dirichlet Poincaré inequality, and $r^{-2} \geq 1$ on $(\rho, 1)$ give

$$\lambda_{\ell,n} \geq \frac{n^2 \pi^2}{(1-\rho)^2} + \nu_{\ell,d}^2 - \frac{1}{4}.$$

Since $\nu_{0,d} = (d-2)/2$, one obtains the strengthened wall

$$N_D(A_{\rho,1}^{(d)}, K^2) = 0 \quad \text{if} \quad K^2 \leq \frac{\pi^2}{(1-\rho)^2} + \frac{(d-2)^2 - 1}{4}. \quad (40)$$

The equality face belongs to the no-mode region because the counting function is strict.

9 Remaining proof obligations

The primary route now consists of the following precise steps.

1. Certify the finite scalar wall list (34) from Proposition 6.5; this completes the first interface-cell milestone.
2. For arbitrary $r < \mu$, prove the shelf-block inequality (39), retaining a quantitative convex suffix reserve. A one-negative-cell/one-drop pairing is not a valid target.
3. Insert the resulting positive shifted-tail theorem into Theorem 5.3. No dimension-dependent angular or optical theorem is then needed.

If the second step fails only for an explicitly controlled family of tails, the exact identity (24) is the appropriate aggregate fallback: the weighted negative part must be compared directly with $\mathcal{B}_d(A)$.

References

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